Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	29/09/2026
Team ID	SWTID-2026-2908
Project Name	FitSense – AI-Powered Fitness Recommendation System
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed FitSense solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Register through form; validate required details; securely store password using bcrypt.js.
FR-2	User Login & Authentication	Login using email and password; issue JWT token for authenticated access.
FR-3	Workout Management	Add, view, update and delete workout records containing workout name, category, duration, calories burned and workout date.
FR-4	Workout Search	Search previous workout records based on relevant workout information.
FR-5	AI Workout Recommendations	Use age, fitness goal, experience level and workout statistics to generate personalized workout plans using Google Gemini.
FR-6	AI Fitness Insights	Provide performance analysis, improvement suggestions, training tips, safety recommendations and motivational advice.
FR-7	API Response	Return application results through REST APIs in JSON format.

Non-functional Requirements:

Following are the non-functional requirements of the proposed FitSense solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide simple REST API interactions and clear JSON responses for user operations.
NFR-2	Security	Use JWT authentication and bcrypt.js password hashing to protect user access and credentials.
NFR-3	Reliability	Maintain consistent workout records and handle invalid or unauthorized requests with appropriate responses.
NFR-4	Performance	Use Node.js and Express.js for efficient API processing and MongoDB for workout data access.
NFR-5	Availability	Keep the backend services and database accessible during normal application operation.
NFR-6	Scalability	Use modular MVC architecture and MongoDB so additional fitness features can be added later.